

Practice Problems

1. Let $A = \begin{bmatrix} 1 & 5 & -2 & 0 \\ -3 & 1 & 9 & -5 \\ 4 & -8 & -1 & 7 \end{bmatrix}$, $p = \begin{bmatrix} 3 \\ -2 \\ 0 \\ -4 \end{bmatrix}$, and $b = \begin{bmatrix} -7 \\ 9 \\ 0 \end{bmatrix}$. It can be shown that p is a solution of $Ax=b$. Use this fact to exhibit b as a specific linear combination of the columns of A .

Solution: Let us first verify that p is a solution of $Ax=b$. From Theorem 3, p must be the solution of the system of equations corresponding to the augmented matrix $[A \ b]$.

$$\begin{aligned}
 & \left[\begin{array}{cccc|c} 1 & 5 & -2 & 0 & -7 \\ -3 & 1 & 9 & -5 & 9 \\ 4 & -8 & -1 & 7 & 0 \end{array} \right] \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + 3 \times \text{Row 1}}]{\text{Row 3} = \text{Row 3} \\ - 4 \times \text{Row 1}} \left[\begin{array}{cccc|c} 1 & 5 & -2 & 0 & -7 \\ 0 & 16 & 3 & -5 & -12 \\ 4 & -8 & -1 & 7 & 0 \end{array} \right] \\
 & \xrightarrow[\substack{\text{Row 3} = \text{Row 3} \\ \times (1/7)}]{\text{Row 3} = \text{Row 3} \\ \text{Row 3} + (1/4) \times \text{Row 2}} \left[\begin{array}{cccc|c} 1 & 5 & -2 & 0 & -7 \\ 0 & 16 & 3 & -5 & -12 \\ 0 & -28 & 7 & 7 & 28 \end{array} \right] \\
 & \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ \times 4}]{\text{Row 2} = \text{Row 2} \\ \text{Row 1} = \text{Row 1} \\ + (2/7) \times \text{Row 3}} \left[\begin{array}{cccc|c} 1 & 5 & -2 & 0 & -7 \\ 0 & 16 & 3 & -5 & -12 \\ 0 & 0 & 7/4 & -14 & 1 \end{array} \right] \\
 & \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ - \text{Row 3} \times \frac{3}{87}}]{\text{Row 2} = \text{Row 2} \\ \text{Row 1} = \text{Row 1} \\ + 5 \times \text{Row 2}} \left[\begin{array}{cccc|c} 1 & 5 & 0 & -2/7 & -41/7 \\ 0 & 16 & 3 & -5 & -12 \\ 0 & 0 & 7 & -1 & 4 \end{array} \right] \\
 & \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ \times (-1/16)}]{\text{Row 2} = \text{Row 2} \\ \text{Row 1} = \text{Row 1} \\ \times 7} \left[\begin{array}{cccc|c} 1 & 5 & 0 & -2/7 & -41/7 \\ 0 & -1 & 0 & 2/7 & 6/7 \\ 0 & 0 & 7 & -1 & 4 \end{array} \right] \\
 & \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ \times (-7)}]{\text{Row 2} = \text{Row 2} \\ \text{Row 1} = \text{Row 1} \\ \times 7} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 8 & -11 \\ 0 & -1 & 0 & 2/7 & 6/7 \\ 0 & 0 & 7 & -1 & 4 \end{array} \right] \\
 & \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ \times (-7)}]{\text{Row 2} = \text{Row 2} \\ \text{Row 1} = \text{Row 1} \\ \times 7} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 8 & -11 \\ 0 & 1 & 0 & -2 & -6 \\ 0 & 0 & 7 & -1 & 4 \end{array} \right]
 \end{aligned}$$